Domains individually

Lean 4 Formalization Companion – Chapter 2

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Section 2.1

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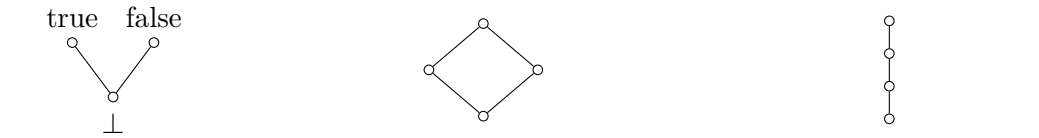
Definition 2.1.1. A set P with a binary relation $\sqsubseteq$ is called a partially ordered set or poset if the following holds for all $x, y, z \in P$:

1. $x \sqsubseteq x$ (Reflexivity)
2. $x \sqsubseteq y \wedge y \sqsubseteq z \Rightarrow x \sqsubseteq z$ (Transitivity)
3. $x \sqsubseteq y \wedge y \sqsubseteq x \Rightarrow x = y$ (Antisymmetry)

If antisymmetry is omitted, the structure is a preorder. The induced equivalence relation is $\sqsubseteq \cap \sqsupseteq$, and passing to equivalence classes gives a canonical poset. A common way to extract order-theoretic content from a relation R is its transitive closure $\bigcup_{n \in \mathbb{N} \setminus \{0\}} R^n$.

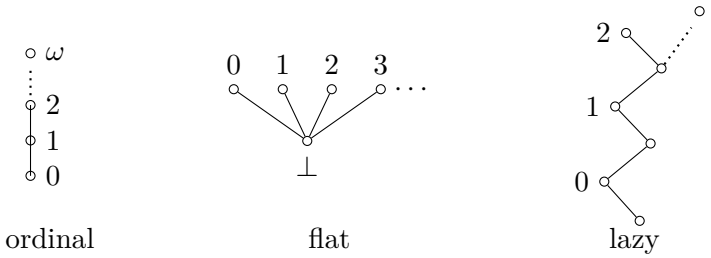
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Proposition 2.1.2. If $\langle P, \sqsubseteq \rangle$ is an ordered set, then so is $P^{op} = \langle P, \sqsupseteq \rangle$.



The flat booleans | The four-element lattice | The four-element chain

Figure 1: A few posets drawn as line diagrams.



ordinal | flat | lazy

Figure 2: Three versions of the natural numbers.

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Definition 2.1.3. Let $(P, \sqsubseteq)$ be an ordered set.

1. A subset A of P is an upper set if $x \in A$ and $y \sqsupseteq x$ imply $y \in A$. We write $\uparrow A$ for the set of elements above some element of A, and abbreviate $\uparrow \{x\}$ as $\uparrow x$. The dual notions are lower set and $\downarrow A$.

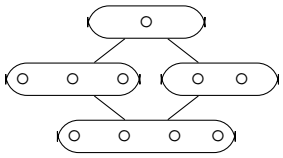


Figure 3: A preorder whose canonical quotient is the four-element lattice.

2. An element $x \in P$ is an upper bound for $A \subseteq P$ if every element of A is below x; we write $A \sqsubseteq x$. The sets of upper and lower bounds are denoted $\mathrm{ub}(A)$ and $\mathrm{lb}(A)$.

3. An element $x \in P$ is maximal if $\uparrow x \cap P = \{x\}$. Minimal elements are defined dually. Minimal upper bounds of A are collected in $\mathrm{mub}(A)$.

4. If all elements of P are below a single element $x \in P$, then x is the largest element. The dually defined least element is also called bottom and is denoted $\bot$.

5. If $A \subseteq P$ has a least upper bound x, then x is the supremum or join of A, written $x = \bigvee A$. Dually, an infimum or meet is written $x = \bigwedge A$.

6. A poset is a $\sqcup$-semilattice, respectively $\sqcap$-semilattice, if each pair has a supremum, respectively infimum. It is a lattice if both operations exist, and complete if suprema and infima exist for all subsets.

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Proposition 2.1.4. Let P be a poset such that the suprema and infima occurring below exist. Let A, B and all A_i be subsets of P. Lean

1. If $A \subseteq B$, then $\bigvee A \sqsubseteq \bigvee B$ and $\bigwedge A \sqsupseteq \bigwedge B$.

2. $\bigvee A = \bigvee(\downarrow A)$ and $\bigwedge A = \bigwedge(\uparrow A)$.

3. If $A = \bigcup_{i \in I} A_i$, then
$$\bigvee A = \bigvee_{i \in I} (\bigvee A_i),$$
and similarly for the infimum.

Proof. For (3), $\bigvee A$ is above every A_i, hence above each $\bigvee A_i$. Conversely, each $a \in A$ belongs to some A_i, so it is below $\bigvee A_i$ and therefore below $\bigvee_i(\bigvee A_i)$. This proves the displayed equality; the infimum statement is dual.

Proposition 2.1.5. Every closure system is a complete lattice with respect to inclusion. Infima are given by intersections; the supremum of a family is the closure of its union. Lean

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Definition 2.1.6. Let P and Q be partially ordered sets. A function $f : P \to Q$ is called monotone if, for all $x, y \in P$, Lean

$$x \sqsubseteq y \quad \Rightarrow \quad f(x) \sqsubseteq f(y).$$

The set $[P \xrightarrow{m} Q]$ of monotone functions is ordered pointwise: $f \sqsubseteq g$ iff $f(x) \sqsubseteq g(x)$ for all $x \in P$.

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Proposition 2.1.7. If L is a complete lattice, then every monotone map $f : L \to L$ has a fixed point. The least fixed point is

$$\bigwedge\{x \in L \mid f(x) \sqsubseteq x\},$$

and the greatest fixed point is

$$\bigvee\{x \in L \mid x \sqsubseteq f(x)\}.$$

Proof. Let

$$A = \{x \in L \mid f(x) \sqsubseteq x\}, \qquad a = \bigwedge A.$$

For every $x \in A$ we have $a \sqsubseteq x$, hence $f(a) \sqsubseteq f(x) \sqsubseteq x$. Taking the infimum over A gives $f(a) \sqsubseteq a$, so $a \in A$. Monotonicity then gives $f(a) \in A$, and therefore $a \sqsubseteq f(a)$. Thus $f(a) = a$, and a is the least fixed point. The greatest fixed point is obtained dually from $\bigvee\{x \in L \mid x \sqsubseteq f(x)\}$.

For lattices, the converse is also true: the existence of fixed points for monotone maps implies completeness. The proof is harder and relies on the Axiom of Choice; see Markowsky.

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Definition 2.1.8. Let P be a poset. A subset A of P is directed if it is nonempty and each pair of elements of A has an upper bound in A. If a directed set A has a supremum, it is denoted by $\bigvee^{\uparrow} A$. Directed lower sets are called ideals; ideals of the form $\downarrow x$ are principal. Dually one obtains filtered sets and filters. Chains are nonempty totally ordered subsets, and subsets isomorphic to the natural numbers are called ω-chains.

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Proposition 2.1.9. Let A be a nonempty subset of a $\sqcup$-semilattice for which $\bigvee A$ exists. Then the join of A can also be written as the directed join

$$\bigvee A = \bigvee \left\{ \bigvee M \;\middle|\; M \subseteq A,\ M \text{ finite and nonempty} \right\}.$$

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Definition 2.1.10. A monotone net in a poset P is a monotone function $\alpha : I \to P$ from a directed set I into P; I is called the index set. If $\alpha : I \to P$ is a monotone net and $\beta : J \to I$ is monotone, with J directed and cofinal in the sense that for every $i \in I$ there is $j \in J$ with $\beta(j) \geq i$, then $\alpha \circ \beta : J \to P$ is called a subnet of α. A monotone net has a supremum in P when the set $\{\alpha(i) \mid i \in I\}$ has a supremum in P.

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Lemma 2.1.11. Let P be a poset and let $\alpha : I \to P$ be a monotone net. Then α has a subnet $\alpha \circ \beta : J \to P$ whose index set J is a lattice and in which every principal ideal is finite.

For the construction, take J to be the set of finite subsets of I. Define $\beta : J \to I$ by induction on cardinality:

$$\beta(\emptyset) = \text{any element of } I,$$

and, for $A \neq \emptyset$,

$$\beta(A) = \text{any upper bound of } A \cup \{\beta(B) \mid B \subset A\}.$$

Then β is monotone and cofinal, hence defines the required subnet.

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Proposition 2.1.12. Let I be directed and let $\alpha : I \times I \to P$ be a monotone net. Assuming the indicated directed suprema exist, the iterated suprema and the diagonal supremum agree:

$$\bigvee\{\alpha(i,j) \mid i,j \in I\} = \bigvee_{i \in I}^{\uparrow} \left(\bigvee_{j \in I}^{\uparrow} \alpha(i,j) \right) = \bigvee_{j \in I}^{\uparrow} \left(\bigvee_{i \in I}^{\uparrow} \alpha(i,j) \right) = \bigvee_{i \in I}^{\uparrow} \alpha(i,i).$$

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Definition 2.1.13. A poset D is called a directed-complete partial order, or dcpo, if every directed subset of D has a supremum. Lean

Examples 2.1.14. Every complete lattice is a dcpo; every finite poset is a dcpo; the natural numbers with the usual order are not a dcpo unless a top element is added. Further examples include point sets of locales ordered by specialization.

Proposition 2.1.15. A partially ordered set D is a dcpo if and only if each chain in D has a supremum. Lean

The proof uses the Axiom of Choice and goes back to a lemma of Iwamura; the cited reference is Markowsky.

Proposition 2.1.16. A pointed poset P is a dcpo if and only if every monotone map $P \to P$ has a least fixed point. Lean

Definition 2.1.17. Let D and E be dcpos. A function $f : D \to E$ is *Scott-continuous* if it is monotone and preserves directed suprema: for every directed subset $A \subseteq D$, Lean

$$f\left(\bigvee^{\uparrow} A\right) = \bigvee^{\uparrow} f(A).$$

The continuous functions from D to E, ordered pointwise, are denoted $[D \to E]$. A function between pointed dcpos is *strict* if it preserves bottom; continuous strict functions are denoted $[D \xrightarrow{\perp!} E]$. The identity on A is written id_A, and the constant function with value x is written c_x.

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Proposition 2.1.18. Let D and E be dcpos. Then $[D \to E]$ is again a dcpo. Directed suprema in $[D \to E]$ are calculated pointwise. Lean

Proof. Let F be a directed collection of continuous functions from D to E. Define

$$g(x) = \bigvee_{f \in F}^{\uparrow} f(x).$$

For a directed set $A \subseteq D$ one computes

$$\begin{aligned} g\left(\bigvee^{\uparrow} A\right) &= \bigvee_{f \in F}^{\uparrow} f\left(\bigvee^{\uparrow} A\right) \\ &= \bigvee_{f \in F}^{\uparrow} \bigvee_{a \in A}^{\uparrow} f(a) && \text{(continuity of } f\text{)} \\ &= \bigvee_{a \in A}^{\uparrow} \bigvee_{f \in F}^{\uparrow} f(a) && \text{(Proposition 2.1.12)} \\ &= \bigvee_{a \in A}^{\uparrow} g(a). \end{aligned}$$

Thus g is continuous.

Theorem 2.1.19. Let D be a pointed dcpo. Lean

1. Every continuous function $f : D \to D$ has a least fixed point, namely

$$\mathrm{fix}(f) = \bigvee_{n \in \mathbb{N}}^{\uparrow} f^n(\perp).$$

2. The assignment

$$\mathrm{fix} : [D \to D] \to D, \qquad f \mapsto \bigvee_{n\in\mathbb{N}}^{\uparrow} f^n(\bot),$$

is continuous.

The first part is the Kleene fixed point theorem: the iterates form an increasing chain, continuity makes the supremum a fixed point, and induction shows it is below every other fixed point. The second part follows because the iteration operators $\mathrm{it}_n(f) = f^n(\bot)$ are continuous and fix is their pointwise supremum.

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Lemma 2.1.20. Let D be a dcpo, let $P \subseteq D$ be an admissible predicate, and let $f : D \to D$ be Scott-continuous. If $f(x)$ satisfies P whenever x satisfies P, then $\mathrm{fix}(f)$ satisfies P. Lean

Lemma 2.1.21. Let D and E be pointed dcpos, let $f : D \to D$ and $g : E \to E$ be continuous, and let $h : D \to E$ be strict and continuous. If the square commutes, $h \circ f = g \circ h$, then Lean

$$\mathrm{fix}(g) = h(\mathrm{fix}(f)).$$

Indeed,

$$\begin{aligned} h(\mathrm{fix}(f)) &= h\left(\bigvee_{n\in\mathbb{N}}^{\uparrow} f^n(\bot)\right) \\ &= \bigvee_{n\in\mathbb{N}}^{\uparrow} h(f^n(\bot)) \\ &= \bigvee_{n\in\mathbb{N}}^{\uparrow} g^n(h(\bot)) \\ &= \mathrm{fix}(g), \end{aligned}$$

using continuity of h, commutativity, and strictness of h.

Section 2.2

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Definition 2.2.1. Let x and y be elements of a dcpo D. We say that x *approximates* y, written $x \ll y$, if for every directed subset $A \subseteq D$,

$$y \sqsubseteq \bigvee^{\uparrow} A \quad \Rightarrow \quad \exists a \in A,\ x \sqsubseteq a.$$

The element x is *compact* if $x \ll x$. For $x, y \in D$ and $A \subseteq D$ the standard notation is

$$\begin{array}{rcl} x \ll y & \iff & x \text{ approximates } y, \\ \Downarrow x & = & \{y \in D \mid y \ll x\}, \\ \Uparrow x & = & \{y \in D \mid x \ll y\}, \\ \Uparrow A & = & \displaystyle\bigcup_{a \in A} \Uparrow a, \\ K(D) & = & \{x \in D \mid x \text{ compact}\}. \end{array}$$

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Proposition 2.2.2. Let D be a dcpo. For all $x, x', y, y' \in D$:

1. $x \ll y \Rightarrow x \sqsubseteq y$.
2. $x' \sqsubseteq x \ll y \sqsubseteq y' \Rightarrow x' \ll y'$.

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Definition 2.2.3. A subset B of a dcpo D is a *basis* for D if, for every element $x \in D$, the set

$$B_x = \Downarrow x \cap B$$

contains a directed subset with supremum x. Elements of B_x are called approximants to x relative to B.

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Proposition 2.2.4. Let D be a dcpo with basis B.

1. For every $x \in D$, the set B_x is directed and
$$x = \bigvee^{\uparrow} B_x.$$
2. B contains $K(D)$.
3. Every superset of B is also a basis for D.

Proof. For (1), the supremum is clear from the definition; directedness follows by comparing any two approximants with a directed subset of B_x whose supremum is x. For (2), if $c \in K(D)$, then from $c = \bigvee^{\uparrow} B_c$ compactness gives some $b \in B_c$ above c, while all elements of B_c are below c, so $b = c$. Part (3) is immediate from the definition.

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Corollary 2.2.5. Let D be a dcpo with basis B.

1. The largest basis for D is D itself.
2. B is the smallest basis for D if and only if $B = K(D)$.

The reverse direction in (2) is not a direct consequence of Proposition 2.2.4; the book leaves it as Exercise 2.3.9(26).

Definition 2.2.6. A dcpo is *continuous*, or a continuous domain, if it has a basis. It is *algebraic*, or an algebraic domain, if it has a basis of compact elements. It is ω-continuous if it has a countable basis, and ω-algebraic if $K(D)$ is a countable basis.

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Proposition 2.2.7.

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1. A dcpo D is continuous if and only if, for every $x \in D$,

$$x = \bigvee^{\uparrow} \Uparrow x.$$

2. A dcpo D is algebraic if and only if, for every $x \in D$,

$$x = \bigvee^{\uparrow} K(D)_x.$$

Definition 2.2.8. A closure system L is called *inductive* if it is closed under directed unions.

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Proposition 2.2.9. Every inductive closure system L is an algebraic lattice. Its compact elements are precisely the finitely generated hulls.

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Proof. If A is the hull of a finite set M and $(B_i)_{i\in I}$ is a directed family of hulls with

$$\bigvee_{i\in I}^{\uparrow} B_i = \bigcup_{i\in I} B_i \supseteq A,$$

then M is contained in some B_i. Hence finitely generated hulls are compact. Conversely, every closed set is the directed union of finitely generated hulls, so these hulls form a basis; Proposition 2.2.4(2) rules out any other compact elements.

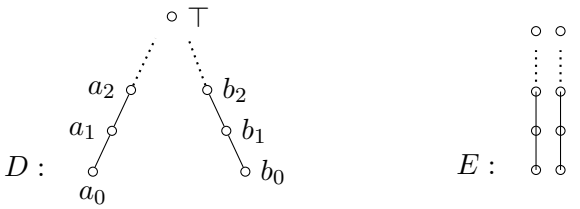


Figure 4: A continuous (E) and a non-continuous (D) dcpo.

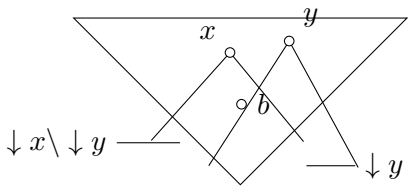


Figure 5: Basis element b witnesses that x is not below y.

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Proposition 2.2.10. Let D be a continuous domain with basis B, and let $x, y \in D$. The following are equivalent:

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$$x \sqsubseteq y, \qquad B_x \subseteq B_y, \qquad B_x \subseteq \downarrow y.$$

Equivalently, if $x \not\sqsubseteq y$, then there exists $b \in B_x$ with $b \not\sqsubseteq y$. Figure 5 depicts this situation.

In the light of Proposition 2.2.10, the dcpo D in Figure 4 is not continuous because the natural would-be basis consisting of the a_i's and b_i's cannot see that

$$\bigvee_{n\in\mathbb{N}}^{\uparrow} a_n = \bigvee_{n\in\mathbb{N}}^{\uparrow} b_n = \top.$$

In a continuous domain this information would already be reflected by the basis. This is why the continuous version should look like E in Figure 4, which is indeed algebraic.

Bases also allow us to express continuity of functions in a form reminiscent of the ϵ-δ definition for real-valued functions.

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Proposition 2.2.11. Let $f : D \to E$ be a map between continuous domains with bases B and C, respectively. Then f is continuous if and only if, for every $x \in D$ and every $e \in C_{f(x)}$, there exists $d \in B_x$ such that

$$f(\uparrow d) \subseteq \uparrow e.$$

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Proof. If f is continuous, then

$$f(x) = f\left(\bigvee^{\uparrow} B_x\right) = \bigvee_{d\in B_x}^{\uparrow} f(d).$$

Since $e \ll f(x)$, some $d \in B_x$ satisfies $e \sqsubseteq f(d)$; monotonicity gives $f(\uparrow d) \subseteq \uparrow e$.

Conversely, first suppose $x \sqsubseteq y$ but $f(x) \not\sqsubseteq f(y)$. By Proposition 2.2.10 choose $e \in C_{f(x)} \setminus \downarrow f(y)$, and then $d \in B_x$ with $f(\uparrow d) \subseteq \uparrow e$. Since $y \in \uparrow d$, this contradicts $e \not\sqsubseteq f(y)$. Thus f is monotone. Now let $A \subseteq D$ be directed with $x = \bigvee^{\uparrow} A$. Monotonicity gives $\bigvee^{\uparrow} f(A) \sqsubseteq f(x)$. If the reverse inequality failed, Proposition 2.2.10 would give $e \in C_{f(x)}$ with $e \not\sqsubseteq \bigvee^{\uparrow} f(A)$; choose $d \in B_x$ with $f(\uparrow d) \subseteq \uparrow e$. Since $d \ll x$, some $a \in A$ lies above d, hence $\bigvee^{\uparrow} f(A) \sqsupseteq f(a) \sqsupseteq f(d) \sqsupseteq e$, a contradiction.

This criterion is followed by a fixed-point result for countable bases: a continuous function need not preserve compactness or the order of approximation, so the naive sequence $\bot, f(\bot), f(f(\bot)), \ldots$ need not consist of basis elements.

Proposition 2.2.12. Let D be a pointed ω-continuous domain with basis B, and let $f : D \to D$ be continuous. Then there exists an ω-chain

$$b_0 \sqsubseteq b_1 \sqsubseteq b_2 \sqsubseteq \cdots$$

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of basis elements satisfying:

1. $b_0 = \bot$;
2. for all $n \in \mathbb{N}$, $b_{n+1} \sqsubseteq f(b_n)$;
3. $$\bigvee_{n\in\mathbb{N}}^{\uparrow} b_n = \mathrm{fix}(f) = \bigvee_{n\in\mathbb{N}}^{\uparrow} f^n(\bot).$$

The book cites Abramsky for the proof.

Proposition 2.2.13. If a dcpo D has a countable basis, then every directed subset of D contains an ω-chain with the same supremum.

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This motivates a parallel ω-chain presentation: an ω-ccpo has suprema of all ω-chains; an ω-continuous map preserves those suprema; and x ω-approximates y if

$$y \sqsubseteq \bigvee_{n\in\mathbb{N}}^{\uparrow} a_n \quad \Rightarrow \quad \exists n \in \mathbb{N},\ x \sqsubseteq a_n.$$

With a countable basis, this version agrees with the directed-set presentation.

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Proposition 2.2.14.

1. Every continuous ω-ccpo is a continuous dcpo.
2. Every algebraic ω-ccpo is an algebraic dcpo.
3. Every ω-continuous map between continuous ω-ccpos is continuous.

Proof. Enumerate a countable basis B as $(b_n)_{n\in\mathbb{N}}$. For a directed subset A, construct an ω-chain $(a_n)_{n\in\mathbb{N}}$ in A that successively dominates the relevant basis elements below members of A. The resulting chain need not be cofinal in A, but it has the same supremum because both dominate the same basis elements. This also identifies ω-approximation with ordinary approximation in a continuous ω-ccpo, yielding (1); part (2) follows from the same argument. For (3), if f is ω-continuous and A is directed, choose such an ω-chain (a_n) in A. Then

$$f\left(\bigvee^{\uparrow} A\right) = f\left(\bigvee_{n\in\mathbb{N}}^{\uparrow} a_n\right) = \bigvee_{n\in\mathbb{N}}^{\uparrow} f(a_n) \sqsubseteq \bigvee^{\uparrow} f(A),$$

while the opposite inequality follows from monotonicity.

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Lemma 2.2.15. Let D be a continuous domain and let $M \subseteq D$ be a finite set whose elements all approximate y. Then there exists $y' \in D$ such that

$$M \ll y' \ll y,$$

meaning $m \ll y'$ for every $m \in M$ and $y' \ll y$. If B is a basis for D, then y' may be chosen from B.

Proof. Define

$$A = \{a \in D \mid \exists a' \in D,\ a \ll a' \ll y\}.$$

The set A is nonempty and directed. Indeed, from $a \ll a' \ll y$ and $b \ll b' \ll y$, directedness of $\Downarrow y$ gives c' above a' and b' with $c' \ll y$; directedness of $\Downarrow c'$ gives c above a and b with $c \ll c'$. Moreover

$$\bigvee^{\uparrow} A = y,$$

because every $y' \ll y$ satisfies $\Downarrow y' \subseteq A$, while all elements of A are below y. For each $m \in M$, choose $a_m \in A$ with $m \sqsubseteq a_m$, and let $a \in A$ be an upper bound of the finite family $(a_m)_{m\in M}$. By definition of A, some a' satisfies $a \ll a' \ll y$, and this a' interpolates between M and y. The basis-refined version is obtained by allowing only basis elements in A.

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Corollary 2.2.16. Let D be a continuous domain with basis B, and let $A \subseteq D$ be directed. If $c \ll \bigvee^{\uparrow} A$, then $c \ll a$ for some $a \in A$. Equivalently,

$$\Downarrow \left(\bigvee^{\uparrow} A \right) = \bigcup_{a \in A} \Downarrow a.$$

Intersecting both sides with the basis gives

$$B_{\bigvee^{\uparrow} A} = \bigcup_{a \in A} B_a.$$

Proposition 2.2.17. Lean

1. If D is continuous and $x, y \in D$, then $x \ll y$ iff for every directed set A with $\bigvee^{\uparrow} A = y$ there exists $a \in A$ with $x \sqsubseteq a$.
2. The order of approximation on a continuous domain is the union of the orders of approximation on all principal ideals.
3. A dcpo is continuous if and only if each principal ideal is continuous.
4. For a continuous domain D,
$$K(D) = \bigcup_{x \in D} K(\downarrow x).$$
5. A dcpo is algebraic if and only if each principal ideal is algebraic.

Proposition 2.2.18. Lean

1. In a continuous domain, minimal upper bounds of finite sets of compact elements are compact.
2. In a complete lattice, the sets $\downarrow x$ are $\sqcup$-sub-semilattices.
3. In a complete lattice, the join of finitely many compact elements is compact.

Corollary 2.2.19. A complete lattice is algebraic if and only if each element is the join of compact elements. Lean

The infimum of compact elements need not be compact, even in an algebraic lattice; Figure 6 gives an example.

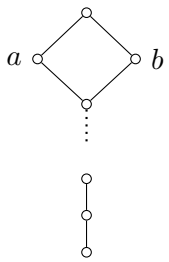


Figure 6: The meet of the compact elements a and b is not compact.

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Definition 2.2.20. An *abstract basis* is a set B equipped with a transitive relation $\prec$ on B such that, for every $x \in B$ and every finite subset $M \subseteq B$,

$$M \prec x \quad \Rightarrow \quad \exists y \in B,\ M \prec y \prec x. \tag{INT}$$

Here $M \prec x$ means $m \prec x$ for every $m \in M$. Concrete bases of continuous domains are examples, with $\prec$ given by the restriction of $\ll$; posets are examples because reflexivity implies interpolation.

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Definition 2.2.21. For a basis $(B, \prec)$, let $\mathrm{Idl}(B)$ be the set of all ideals ordered by inclusion. This dcpo is called the *ideal completion* of B. Let Lean

$$i : B \to \mathrm{Idl}(B)$$

map $x \in B$ to $\downarrow x$. When the relation should be explicit, write $\mathrm{Idl}(B, \prec)$.

Proposition 2.2.22. Let $(B, \prec)$ be an abstract basis. Lean

1. The ideal completion $\mathrm{Idl}(B)$ is a dcpo.
2. For ideals $A, A' \in \mathrm{Idl}(B)$, $A \ll A'$ iff there are $x \prec y$ in B such that

   $$A \subseteq i(x) \subseteq i(y) \subseteq A',$$

   iff there is $x \in A'$ with $A \subseteq i(x)$.
3. $\mathrm{Idl}(B)$ is a continuous domain, with basis $i(B)$.
4. If $\prec$ is reflexive, then $\mathrm{Idl}(B)$ is algebraic.
5. If $(B, \prec)$ is a poset, then B, $K(\mathrm{Idl}(B))$, and $i(B)$ are all isomorphic.

Proof. Directed unions of ideals are ideals. Roundness gives

$$A = \bigcup_{x \in A} \downarrow x,$$

and this union is directed because A is directed. This proves the way-below characterization and continuity. Reflexivity gives algebraicity, and the final clause follows from uniqueness of the basis of compact elements.

Proposition 2.2.23. For abstract bases B and B', ideal completion preserves products: Lean

$$\mathrm{Idl}(B \times B') \cong \mathrm{Idl}(B) \times \mathrm{Idl}(B').$$

The completion also has a weak universal property, stated next.

Proposition 2.2.24. Let $(B, \prec)$ be an abstract basis and let D be a dcpo. For every monotone function $f : B \to D$ there is a largest continuous function Lean

$$\widehat{f} : \mathrm{Idl}(B) \to D$$

such that $\widehat{f} \circ i \sqsubseteq f$. It is given by

$$\widehat{f}(A) = \bigvee^{\uparrow} f(A).$$

The assignment $f \mapsto \widehat{f}$ is Scott-continuous from $[B \xrightarrow{m} D]$ to $[\mathrm{Idl}(B) \to D]$. If $\prec$ is reflexive, then $\widehat{f} \circ i = f$.

Proof. For a directed family $(A_i)_{i\in I}$ of ideals, associativity of directed suprema gives

$$\begin{aligned}\widehat{f}\left(\bigvee^{\uparrow}_{i\in I} A_i\right) &= \widehat{f}\left(\bigcup_{i\in I} A_i\right)\\ &= \bigvee^{\uparrow}\{f(x) \mid x \in \bigcup_{i\in I} A_i\}\\ &= \bigvee^{\uparrow}_{i\in I}\bigvee^{\uparrow}\{f(x) \mid x \in A_i\}\\ &= \bigvee^{\uparrow}_{i\in I} \widehat{f}(A_i).\end{aligned}$$

Monotonicity of f makes $f(x)$ an upper bound for $f(\downarrow x)$, so $\widehat{f}\circ i \sqsubseteq f$. If $g : \mathrm{Idl}(B) \to D$ is continuous and $g \circ i \sqsubseteq f$, then

$$g(A) = g\left(\bigcup_{x\in A} \downarrow x\right) = \bigvee^{\uparrow}_{x\in A} g(i(x)) \sqsubseteq \bigvee^{\uparrow}_{x\in A} f(x) = \widehat{f}(A).$$

The continuity of $f \mapsto \widehat{f}$ follows by interchanging directed suprema. If $\prec$ is reflexive, then $\widehat{f}(i(x)) = \bigvee^{\uparrow} f(\downarrow x) = f(x)$.

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Proposition 2.2.25. Let D be a continuous domain with basis B. Viewing $(B, \ll)$ as an abstract basis: Lean

1. $\mathrm{Idl}(B)$ is isomorphic to D. The isomorphism $\sigma : \mathrm{Idl}(B) \to D$ is the extension $\widehat{e}$ of the embedding $e : B \hookrightarrow D$, and its inverse β maps $x \in D$ to B_x.
2. For every dcpo E and continuous function $f : D \to E$,

   $$f = \widehat{g} \circ \beta,$$

   where g is the restriction of f to B.

Proof. Since $x = \bigvee^{\uparrow} B_x$ in a continuous domain, $\sigma\circ\beta = \mathrm{id}_D$. Conversely, if $c \in B$ approximates $\bigvee^{\uparrow} A$ for an ideal A in $(B, \ll)$, interpolate

$$c \ll d \ll \bigvee^{\uparrow} A.$$

By the definition of $\ll$, some $a \in A$ lies above d, hence $c \ll a$ and $c \in A$. Finally,

$$f(x) = f\left(\bigvee^{\uparrow} B_x\right) = \bigvee^{\uparrow} f(B_x) = \widehat{g}(B_x) = \widehat{g}(\beta(x)).$$

Corollary 2.2.26. A continuous function from a continuous domain D to a dcpo E is completely determined by its behavior on a basis of D. Lean

The preceding result reconstructs continuous domains and continuous maps from bases. Since continuous functions need not preserve $\ll$, the basis-only formulation uses relations rather than functions.

Definition 2.2.27. A relation R between abstract bases B and C is *approximable* if: Lean

1. for $x \in B$ and $y, y' \in C$, xRy and $y' \prec y$ imply xRy';
2. for $x \in B$ and finite $M \subseteq C$, if xRy for every $y \in M$, then there is $z \in C$ such that xRz and $M \prec z$;
3. for $x, x' \in B$ and $y \in C$, $x \prec x'$ and xRy imply $x'Ry$;
4. for $x \in B$ and $y \in C$, xRy implies that there exists $z \in B$ with $z \prec x$ and zRy.

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Theorem 2.2.28. The category of abstract bases and approximable relations is equivalent to **CONT**, the category of continuous dcpos and continuous maps. Lean

Theorem 2.2.29. The category of preorders and approximable relations is equivalent to **ALG**, the category of algebraic dcpos and continuous maps. Lean

Section 2.3

Definition 2.3.1. Let D be a dcpo. A subset $A \subseteq D$ is *Scott-closed* if it is a lower set and is closed under suprema of directed subsets. Complements of Scott-closed sets are *Scott-open*; they form the Scott topology σD. Lean

Write $\mathrm{Cl}(A)$ for the smallest Scott-closed set containing A, and $\mathrm{Int}(A)$ for the open kernel of A. Every Scott-open set is an upper set. Equivalently, a Scott-open set O is an upper set such that every directed set whose supremum lies in O intersects O.

Proposition 2.3.2. Let D be a dcpo. Lean

1. For $x, y \in D$, the following are equivalent:
   (a) $x \sqsubseteq y$;
   (b) every Scott-open set containing x also contains y;
   (c) $x \in \mathrm{Cl}(\{y\})$.
2. The Scott topology satisfies the T_0 separation axiom.
3. $(D, \sigma D)$ is Hausdorff, i.e. T_2, if and only if the order on D is trivial.

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Proposition 2.3.3. Let A be a directed set in a dcpo D. An element $x \in D$ is the supremum of A if and only if x is an upper bound for A and every Scott-neighborhood of x contains an element of A. Lean

Proof. The Scott-closed set

$$\downarrow \left(\bigvee^{\uparrow} A \right)$$

separates the supremum from all other upper bounds of A.

Proposition 2.3.4. For dcpos D and E, a function $f : D \to E$ is Scott-continuous if and only if it is topologically continuous with respect to the Scott topologies. Lean

Proof. If f is Scott-continuous and $O \subseteq E$ is Scott-open, then $f^{-1}(O)$ is an upper set. If $x = \bigvee^{\uparrow}_{i \in I} x_i$ and $f(x) \in O$, then

$$f(x) = f\left(\bigvee_{i \in I}^{\uparrow} x_i \right) = \bigvee_{i \in I}^{\uparrow} f(x_i) \in O,$$

so some x_i lies in $f^{-1}(O)$.

Conversely, assume f is topologically continuous. Monotonicity follows by applying inverse images to principal Scott-closed sets. If $A \subseteq D$ is directed, then the inverse image of

$$\downarrow \left(\bigvee_{a \in A}^{\uparrow} f(a) \right)$$

is Scott-closed and contains A, hence contains $\bigvee^{\uparrow} A$. Together with monotonicity, this shows that f preserves directed suprema.

Proposition 2.3.5. Let D be a continuous domain. For all $x, y \in D$, the following are equivalent: Lean

1. $x \ll y$;
2. $y \in \mathrm{Int}(\Uparrow x)$;
3. $y \in \Uparrow x$.

Of course, (1) and (3) are equivalent in every dcpo by definition of $\Uparrow x$.

Proposition 2.3.6. Let D be a continuous domain with basis B. A subset $O \subseteq D$ is Scott-open iff it can be described in either of the following ways: Lean

1. $$O = \bigcup_{x \in O} \Uparrow x;$$
2. $$O = \bigcup_{x \in O \cap B} \Uparrow x.$$

Thus every open set is supported by its members from the basis.

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Proposition 2.3.7. Let $(B, \prec)$ be an abstract basis and let $M \subseteq B$. Then Lean

$$\{A \in \mathrm{Idl}(B) \mid M \cap A \neq \emptyset\}$$

is Scott-open in $\mathrm{Idl}(B)$, and every Scott-open set of $\mathrm{Idl}(B)$ is of this form.

This identifies Scott-open properties of the completion with observable hits of basis elements.

Lemma 2.3.8. Every Scott-open set in a continuous domain is a union of Scott-open filters. Lean

Proof. Let $x \in O$ with O Scott-open. By Proposition 2.3.6 choose $y \in O$ with $y \ll x$. Repeated interpolation gives a sequence

$$y \ll \cdots \ll y_n \ll \cdots \ll y_1 \ll x.$$

The union of the filters $\Uparrow y_n$ is a Scott-open filter containing x and contained in O.

This completes the chapter's setup for formulating domain theory through the lattice of Scott-open sets.